

GEOMETRY_HANDOUT 02_EXPLANATORY NOTES

Solutions for questions 1 to 15:

1. Drop perpendiculars PA and QB onto RS.

Let PA = QB = h.

$\therefore AB = 10$.

Let AS = x $\Rightarrow$ BR = 21 - x.

In triangle PAS, $h^2 = 10^2 - x^2$

In triangle QBR, $h^2 = (17^2 - (21 - x)^2)$

$$100 - x^2 = 289 - (21 - x)^2$$

$$100 - x^2 = 289 - 21^2 - x^2 + 42x$$

$$x = 6.$$

$$h^2 = 100 - 6^2 = 64$$

$$h = 8.$$

Alternatively, since 17 and 10 are both part of Pythagorean triples, by trial and error, we can arrive at h = 8.

Ans : (8)

2. Let E be the point of intersection of the diagonals.
AE = EC = 4 cm. (Since the diagonals in a parallelogram bisect each other)

In triangle ABC, E is the midpoint of AC.

By Apollonius theorem, $AB^2 + BC^2 = 2(AE^2 + BE^2)$

$$36 + 64 = 2(16 + BE^2)$$

$$BE = \sqrt{34}$$

$$BD = 2BE = 2\sqrt{34} \text{ cm.}$$

Note: In any parallelogram, $d_1^2 + d_2^2 = 2(a^2 + b^2)$ where d_1 , d_2 are the lengths of the diagonals and a, b are the lengths of the sides of the parallelogram.

Ans : ($2\sqrt{34}$ cm)

3. The greatest distance between any two points on the rhombus is equal to the length of the longer diagonal. It is $\sqrt{3}$ times the length of the shorter diagonal.
Hence $2\sqrt{3}$ cm is the length of the longer diagonal.

$$\text{Side of the rhombus} = \sqrt{\frac{(d_1^2 + d_2^2)}{4}} = \sqrt{\frac{(2\sqrt{3})^2 + (2)^2}{4}} = 2 \text{ cm.}$$

Ans : (2 cm)

4. The number of diagonals in the polygon = $\frac{n(n-3)}{2}$
 $= \frac{20(17)}{2} = 170$

Ans : (170).

5. $\frac{n(n-3)}{2} = 35$

$$\Rightarrow n^2 - 3n - 70 = 0$$

$$(n-10)(n+7) = 0.$$

Since 'n' is positive, n = 10.

Measure of each exterior angle of the regular polygon

$$= \frac{360^\circ}{n} = \frac{360^\circ}{10} = 36^\circ.$$

Ans : (36°)

6. Given that, $\frac{3(360^\circ)}{n} = \frac{(180^\circ)(3)}{5}$

$$\Rightarrow n = 10.$$

$$\text{Each interior angle} = 180^\circ - \left(\frac{360^\circ}{n}\right) = 180^\circ - \left(\frac{360^\circ}{10}\right) = 144^\circ.$$

Ans : (144°)

7. Let O be the centre of the regular hexagon. There are six equilateral triangles AOB, BOC, COD, DOE, EOF and FOA.
OB = BC = 10 cm and OE = DE = 10 cm.

$$\angle BOE = \angle BOC + \angle COD + \angle DOE = 60^\circ + 60^\circ + 60^\circ = 180^\circ.$$

BOE is a straight line.

$$\therefore BE = BO + OE = 20 \text{ cm.}$$

$$\text{In triangle BDE, } BD^2 = BE^2 - ED^2 = 20^2 - 10^2 = 300$$

$$\text{Hence, } BD = 10\sqrt{3} \text{ cm}$$

Ans : ($10\sqrt{3}$ cm)

8. OA = OB = OC = radius. Hence, triangles AOB and BOC are isosceles.

$$\therefore \angle OAB = \angle OBA = \frac{180^\circ - 50^\circ}{2} = 65^\circ$$

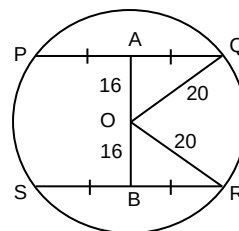
$$\text{Similarly, } \angle OBC = \angle OCB = \frac{180^\circ - 50^\circ}{2} = 65^\circ$$

$$\angle ABC = \angle OBA + \angle OBC = 130^\circ.$$

Alternatively, we can also apply the central angle theorem.

Ans : (130°)

9. Equal chords that are parallel to each other are equidistant from the centre.



$$OA = OB = 16 \text{ cm.}$$

$$OR = 20 \text{ cm}$$

$$\angle OBR = 90^\circ \text{ (Radius bisects the chord perpendicularly)}$$

$$RB^2 = OR^2 - OB^2 = 20^2 - 16^2 = 12^2$$

$$RB = 12.$$

$$\therefore \text{The length of each chord} = RS = 2(RB) = 24.$$

Ans : (24 cm)

10. We know that, $PA \times PB = PT^2$ (Power of a point)

$$8 \times PB = 12 \times 12$$

$$PB = 18.$$

$$AB = PB - PA = 18 - 8 = 10.$$

Ans : (10 cm)

11. OT = OX = OY = 10 = radius.

$$\angle TOA = \angle TOY = 60^\circ. \text{ TOA is a } 30^\circ - 60^\circ - 90^\circ \text{ triangle.}$$

In any such triangle, the ratio of the lengths of the sides opposite 30° , 60° and 90° is $1 : \sqrt{3} : 2$.

$$AO = 2OT = 20$$

$$AX = AO + OX = 20 + 10 = 30.$$

Ans : (30 cm)

12. $\angle BAC = \angle BCY = 70^\circ$ (Alternate segment theorem)

Using central angle theorem, $\angle BOC = 2(\angle BAC)$

$$= 2(70^\circ) = 140^\circ.$$

Ans : (140°)

13. $\angle EAC = 180^\circ - 80^\circ = 100^\circ$

$\angle EAC + \angle CDE = 180^\circ$ (Sum of the opposite angles in a cyclic quadrilateral is 180°).

$$\angle CDE = 80^\circ$$

$$\angle CDE + \angle CBE = 180^\circ \text{ (Cyclic quadrilateral)}$$

$$\angle CBE = 100^\circ.$$

Ans : (100°)

14. The length of each direct common tangent

$$= \sqrt{(\text{Distance between the centres})^2 - (R_1 - R_2)^2}$$

$$= \sqrt{20^2 - (15 - 3)^2} = \sqrt{256} = 16.$$

Ans : (16 cm)

15. $\angle AOB = 2(\angle ACB)$

$$\therefore \angle ACB = 15^\circ$$

$$\text{In triangle ABC, } \angle ABC = 180^\circ - 40^\circ - 15^\circ = 125^\circ.$$

Ans : (125°)

ADDITIONAL QUESTIONS FOR PRACTICE

Solutions for questions 1 to 5:

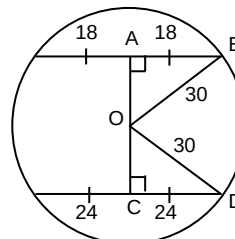
1. $BD = OB + OD$
 $OA \times OC = 9 \times 16 = OB \times OD$ (Power of a point)
 Product of OB and OD is constant.
 $\therefore$ Their sum, i.e., BD is minimum when both are equal.
 The minimum value of BD occurs when $OB = OD$
 $= \sqrt{9 \times 16}$ i.e., 12.
 Hence, minimum value of BD = 24. Ans : (24)

2. $\angle ADC = \angle PAC = 70^\circ$ (Alternate segment theorem)
 $\angle ABC + \angle ADC = 180^\circ$ (Opposite angles in a cyclic quadrilateral are supplementary)
 $\angle ABC = 180^\circ - \angle ADC = 180^\circ - 70^\circ = 110^\circ$.
Ans : (110°)

3. An external angle of a cyclic quadrilateral equals the internal angle of the quadrilateral opposite to it.
 External angle at P = $\angle QRS$ and
 External angle at R = $\angle QPS$
 $\angle QRS : \angle QPS = 2 : 3$
 $\angle QRS + \angle QPS = 180^\circ$ (Opposite angles in a cyclic quadrilateral are supplementary)
 $2x + 3x = 180 \Rightarrow 2x = 72^\circ$. Ans : (72°)

4. $n \left(\frac{n-3}{2} \right) = \left(\frac{3}{100} \right) (180)(n-2)$
 $10n^2 - 30n = 108n - 216$
 $10n^2 - 138n + 216 = 0$
 $10n^2 - 120n - 18n + 216 = 0$
 $n = 12$ or 1.8
 n has to be an integer. Hence, $n = 12$. Ans : (12)

5. The perpendicular drawn from the centre of the circle to the chord, bisects the chord.
 Let A and C be the midpoints of the chords whose lengths are 36 cm and 48 cm respectively. Let B and D be one of the endpoints of the chords whose lengths are 36 cm and 48 cm respectively.



$$\begin{aligned}
 OA^2 &= OB^2 - AB^2 = 30^2 - 18^2 = 576 \\
 \Rightarrow OA &= 24 \\
 OC^2 &= OD^2 - CD^2 = 30^2 - 24^2 = 324 \\
 \Rightarrow OC &= 18 \\
 AC &= OA + OC = 24 + 18 \\
 &= 42 \text{ cm.}
 \end{aligned}$$

Ans : (42 cm)